

Optimising Peer-Assessment Reliability in Capstone Software Projects through Algorithmic Weighting and Qualitative Red-Flagging

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1 Motivation and Background

1.1 The Problem of Individual Assessment in Team Projects

Team-based capstone projects occupy a central role across computer science curricula worldwide, valued for developing the communication, leadership, and engineering skills that employers consistently identify as graduate priorities [4]. Yet the very features that make them pedagogically valuable—shared ownership, interdependent tasks, emergent team dynamics—make individual assessment difficult. A recent multi-institutional survey found that while instructor evaluation was rated as the most influential factor in final grades by 86% of respondents, peer evaluation showed a far more mixed picture: 41% of instructors rated it as “not at all” significant, and many reported that students routinely awarded uniformly high marks regardless of actual performance, or were unwilling to flag underperforming teammates for fear of retaliation [4].

In COMPSCI 399 at the University of Auckland, each student is given a fixed pool of points—ten times the team size—to distribute among their teammates including themselves. Self-scores are excluded when computing each student’s Individual Weighting Factor (IWF): the sum of peer scores received, divided by the number of peers. This factor scales the shared team grade into an individual mark. The intention is sound: a student rated highly by their peers receives a higher grade than one rated poorly, without any student being able to inflate their own score.

In practice, however, the system has significant vulnerabilities that this research seeks to address. Because self-scores are excluded but the denominator remains fixed at $N - 1$, a student who awards all their points to a single teammate (or distributes them strategically rather than honestly) can substantially distort the IWF of others without any check on the quality or consistency of their own ratings. Furthermore, in a cohort of approximately 250+ students, it is infeasible for teaching staff to manually audit every team’s score distribution for signs of collusion, strategic manipulation, or the marginalisation of individual team members. The result is that the current system is trusted by default, despite there being no mechanism to verify its integrity.

Peer assessment is widely deployed as the primary mechanism for distinguishing individual performance in team projects, yet the systems currently in use are so vulnerable to strategic behaviour that many instructors effectively disregard them. This research investigates whether the reliability of peer assessment can be substantially improved through the application of computationally principled weighting algorithms, and whether qualitative data from student journals can complement numerical scores to surface cases that arithmetic alone cannot detect.

A critical design principle underpins both strands of this work: **no algorithmic output should ever replace instructor judgment**. The goal is not to automate grading, but to provide teaching

staff with a prioritised list of cases that warrant closer attention—cases that would otherwise go unnoticed in a cohort of 250 or more students. A human must always review and confirm any action taken on the basis of the system’s output. This principle is not merely a safeguard against false positives; it is a reflection of the ethical obligation that grading decisions remain the responsibility of the instructor, not the algorithm.

1.2 The Individual Weighting Factor Model and Its Limitations

Where peer assessment does influence grading, a common framework is the *Individual Weighting Factor* (IWF) model introduced by Kaufman, Felder and Fuller [6]. In COMPSCI 399, each member of a team of N students distributes $10 \times N$ points across N teammates, including themselves. Although self-scores are excluded, a student’s IWF is computed as the sum of the peer scores they received, divided by $N - 1$:

$$\text{IWF}_i = \frac{\sum_{j \neq i} s_{ji}}{N - 1} \quad (1)$$

where s_{ji} is the score awarded to student i by peer j . The individual grade is then the product of the shared team grade and this factor. An IWF of 1 indicates a student contributed exactly their expected share; values above 1 are rewarded with a grade uplift, and values below 1 attract a penalty.

Despite its elegance, this formulation rests on a critical assumption—that all raters are equally objective and competent—which is rarely satisfied in practice. Three distinct failure modes arise from this assumption.

Collusion through uniform inflation. If an entire team agrees, tacitly or explicitly, to award each other maximum scores, every student’s IWF collapses to 1.0. No differentiation is produced, and the mechanism is entirely bypassed. Hooshangi et al. [4] confirmed this pattern directly from instructor interviews, with one respondent explicitly noting that students could “game the system” by informally agreeing to rate one another positively. This is not a theoretical concern: it is a commonly observed classroom behaviour.

Self-sacrificial grading. Although self-scores are excluded from the IWF calculation, students in COMPSCI 399 are still able to include themselves when distributing their $10 \times N$ points. This creates a direct exploit. A student who awards themselves a score of 0 has their self-score silently discarded, but the remaining points they awarded to teammates are counted in full. Because the denominator is fixed at $N - 1$ regardless of how many points were actually distributed to peers, awarding oneself 0 effectively injects extra points into the pool available to teammates without any penalty to the giver.

Consider a team of three students, where each student has 30 points to distribute. Under honest reporting, a student might award 10 to each teammate and 10 to themselves. Their self-score is discarded, so each teammate receives a score of 10, yielding an IWF of $\frac{10+10}{2} = 10$ for each student. Now suppose all three students coordinate to award themselves 0 and distribute all 30 points evenly to their two teammates: 15 each. Every student now receives scores of 15 and 15 from their peers, giving an IWF of $\frac{15+15}{2} = 15$ for each member — well above the honest baseline of 10. If the whole team agrees to this manoeuvre, every member receives a grade uplift with no corresponding increase in contribution, entirely bypassing the intent of the system.

Rater unreliability. Even absent deliberate manipulation, peer scores vary in quality. Hall and Buzwell [2] draw a critical distinction between *voluntary* free-riders—students who deliberately undercontribute—and *involuntary* free-riders—students who are excluded from team processes due to social dynamics, language barriers, or the consolidation of status hierarchies in the early stages of a project. A student who has been marginalised may lack both the opportunity and the insight to assess their peers’ technical contributions accurately. Under the simple average model, that student’s low-quality ratings are counted with an identical weight to those of the team’s most engaged contributor.

1.3 Research Novelty and Approach

This research proposes a two-strand intervention. The first strand is **algorithmic**: it applies and benchmarks a family of computationally principled weighting models against the University of Auckland’s historical COMPSCI 399 capstone peer-score dataset, testing each model’s resistance to the three failure modes described above. The second strand is **qualitative**: it investigates whether sentiment analysis applied to students’ periodic journal entries can identify cases where a student’s written affect periodically diverges from their received peer scores, flagging potential cases of social marginalisation or coerced grading for instructor review.

The algorithmic family evaluated in this research spans three design philosophies. The **normalisation model**—WebPA [8]—scales each student’s IWF relative to the team’s internal distribution, preserving grade neutrality at the cohort level.

The central iterative contribution is the application of **PeerRank** [12] to this context. Unlike the simple average, PeerRank introduces a credibility weighting inspired by Google’s PageRank algorithm [9]: a student’s grading power is iteratively scaled by their own peer-evaluated standing. A student rated poorly by their team has their votes dampened in subsequent iterations, creating a structural incentive for honest grading. Walsh [12] demonstrated that PeerRank reduces grade prediction error by a factor of two or more compared to averaging, and that it retains meaningful robustness to grading bias of up to $\pm 25\%$ without significant degradation. This project will test whether that robustness holds in the sparse 4-to-6-node graphs characteristic of capstone teams, where a single outlier rater has proportionally greater influence than in the larger cohorts Walsh evaluated.

This research additionally proposes **PeerHITS**, a novel dual-score variant inspired by Kleinberg’s HITS algorithm [7]. Where PeerRank assigns each student a single grade, PeerHITS maintains separate authority and hub scores per student, separating contribution quality from assessment quality. This is described in detail in Section 3.3.

On the qualitative side, Friess and Goupee [1] demonstrated that continuous weekly peer evaluation in engineering capstone courses produces more informative and actionable signals than end-of-project assessment alone, and that students themselves judge such systems to be fairer. Their finding supports the use of longitudinal journal data as a running record of team dynamics rather than a static end-point snapshot. Hundhausen et al. [5] further showed that objective Git repository metrics—commits, lines changed, and story completions—are significantly positively correlated with peer evaluation ratings across 23 undergraduate software teams, providing an independent empirical baseline against which the accuracy of different IWF models can be assessed.

Together, these threads motivate a research design that is, to the author’s knowledge, the first to apply iterative and normalisation peer-assessment models to a longitudinal institutional dataset of capstone peer scores, to cross-validate those models against Git contribution metrics, and to augment the numerical analysis with a qualitative red-flagging pipeline grounded in journal sentiment.

1.4 Related Work

Peer assessment in higher education has attracted sustained research attention, but the focus has predominantly been pedagogical: how peer review affects learning outcomes, student engagement, and feedback quality. Far less attention has been paid to the computational question of how peer ratings should be *combined* to produce the most accurate and manipulation-resistant individual grade adjustment. The following review organises the computational literature into three algorithmic families, each representing a distinct design philosophy, and concludes with work on objective contribution metrics that complements the purely subjective peer-rating approach.

Normalisation models. The WebPA framework [8], developed at Loughborough University and widely adopted in UK higher education, computes a Peer Assessment Factor (PAF) for each student by dividing the total marks they received from peers by the average total marks received across the team. This factor scales the shared group grade into an individual mark, preserving grade neutrality: the weighted marks still sum to the original team grade. WebPA’s primary strength is simplicity and transparency under honest reporting. Its critical weakness is that the PAF is computed from raw scores with no correction for rating strategy: if all members of a team submit identical scores, every PAF equals 1.0, and the mechanism fails entirely to differentiate individual contributions.

Iterative graph models. The PeerRank method [12] represents a fundamentally different design philosophy, treating peer assessment as a network problem. Each student is a node in a directed graph; the edge weight from student j to student i represents j ’s evaluation of i . PeerRank defines each student’s grade as a fixed-point of an equation in which the weight given to j ’s evaluations is proportional to j ’s own grade. The algorithm iterates to convergence, rewarding students whose evaluations align with consensus and penalising those whose assessments deviate from it. Because a student with a low grade has their votes dampened, PeerRank is substantially more resistant to random or dishonest grading than simple averaging. Walsh [12] showed that the method halves grade prediction error relative to averaging and retains robustness to systematic bias of up to 25%. A limitation in the capstone context is that PeerRank was designed and evaluated on large MOOC cohorts where students grade independent individual submissions; its behaviour on the 4-to-6-node directed graphs typical of small capstone teams assessing shared contributions has not been studied.

PeerRank draws its inspiration from PageRank [9], in which a web page’s importance is determined by the importance of the pages that link to it. A closely related but structurally distinct algorithm is HITS (Hyperlink-Induced Topic Search) [7], which assigns each node in a directed graph *two* scores rather than one: an **authority score**, measuring how much the node is endorsed by others, and a **hub score**, measuring how reliably the node endorses high-quality others. These scores reinforce each other iteratively via two alternating update operations: the authority score of a node is set to the sum of the hub scores of all nodes pointing to it, and the hub score is set to the sum of the authority scores of all nodes it points to. Both vectors are L_2 -normalised after each iteration, and the process converges to the principal eigenvectors of $A^T A$ and AA^T respectively, where A is the adjacency matrix of the graph [7].

This dual-score structure has a natural and previously unexplored mapping onto the peer assessment problem. A student’s authority score would capture the quality of their *contribution* — how highly, and by whom, they are assessed. A student’s hub score would capture the quality of their *assessments* — whether the peers they rate highly are themselves rated highly by others. A student who is a strong contributor but a poor assessor (for example, a marginalised team member who lacks full visibility of others’ work) would receive a high authority score but a low hub score.

Conversely, a student who contributes little but calibrates their assessments accurately would receive a low authority score but a high hub score. This separation — which PeerRank collapses into a single grade — provides a richer diagnostic signal. This research proposes **PeerHITS** as a candidate extension to the model family, and will evaluate whether the separation of authority and hub scores produces more informative instructor-facing diagnostics than PeerRank.

Objective contribution metrics. A parallel strand of research bypasses subjective ratings entirely and instead measures individual contribution through observable artefacts of the software development process. Hamer et al. [3] demonstrated that Git metrics—commits, merges, and code churn—can be used to identify contribution inequality within student teams, with Gini Index values indicating high disparity across many projects. Hundhausen et al. [5] showed that Git contributions and team chat activity are significantly positively correlated with peer evaluation scores in undergraduate software teams, providing empirical support for using these metrics as an independent validation baseline. Critically, they also found that the equality of a team’s GitHub contributions (measured by the Gini Index) was positively correlated with average peer evaluation ratings and negatively correlated with the variance in those ratings—suggesting that highly unequal commit distributions are detectable through peer scores, and vice versa. When a team’s Git Gini Index is high but their peer scores are uniformly equal, this divergence is itself a strong signal of collusion or rating inaccuracy.

Table 1 summarises all four models evaluated in this research across the vulnerability dimensions most relevant to this context. **Collusion Resistant** indicates whether the model produces meaningfully differentiated IWFs when an entire team submits identical scores. **Zero-Self Resistant** indicates whether the model is immune to the exploit of a student awarding themselves zero points to inflate teammates’ IWFs. **Rater Reliability** indicates whether the model down-weights evaluations from students who are themselves rated as low contributors, reducing the influence of uninformed or biased raters. **Accuracy** reflects how closely the model’s IWF vector approximates the true (unobservable) contribution vector $\mathbf{t} = (t_1, \dots, t_N)$ under honest reporting, assessed operationally via two proxies: (i) the reduction in root mean squared error (RMSE) between the model’s output and synthetic ground-truth scores in controlled simulations, following Walsh’s methodology [12], and (ii) the Pearson correlation between the model’s IWF vector and Git-derived relative contribution shares in the historical dataset. *Low* indicates no correction for rater quality differences; *Medium* indicates scale-bias reduction without credibility weighting; *High* indicates iterative credibility weighting that has been shown to converge toward the true vector [12].

Table 1: Comparison of peer-assessment models across key vulnerability dimensions. PeerRank and PeerHITS are not zero-self resistant because they operate on the same post-exclusion peer-score matrix as the baseline; the zero-self exploit injects surplus points before any model processes the data.

Model	Collusion Resistant	Zero-Self Resistant	Rater Reliability	Accuracy
Simple Average (Baseline)	No	No	No	Low
WebPA	No	Yes	Partial	Medium
PeerRank	Yes	No	Yes	High
PeerHITS (Proposed)	Yes	No	Yes	High

2 Research Questions and Goals

The overarching goal of this research is to determine whether algorithmically weighted peer-assessment models, augmented by qualitative sentiment signals, can produce more reliable and manipulation-resistant Individual Weighting Factors than the simple average currently deployed in capstone courses. This goal decomposes into four specific research questions.

1. **RQ1 — Resistance to Strategic Manipulation:** How effectively do the normalisation model (WebPA) and the iterative credibility models (PeerRank, PeerHITS) resist collusion through uniform inflation and zero-self assignment, compared to the simple average baseline, when applied to peer-score data from teams of 4 to 6 students?

This question targets the most practically urgent problem: the ease with which small, cohesive teams can render the current system ineffective. It will be evaluated through both historical data analysis and synthetic attack injection.

2. **RQ2 — Convergence Stability in Small Graphs:** How sensitive is PeerRank’s fixed-point solution to single outlier raters in the sparse 4-to-6-node directed graphs typical of capstone teams, and under what conditions does the algorithm fail to converge or produce unstable IWFs?

This question addresses a theoretical limitation of PeerRank that has not been studied in small-group settings. Walsh’s original evaluation used large MOOC cohorts where the influence of any single rater is diluted. In a team of four, one anomalous rater represents 25% of the grading graph, and their influence on the fixed-point is proportionally much greater.

3. **RQ3 — Delta-Analysis as a Collusion Detector:** Can the divergence between IWFs produced by different algorithmic models (the “Delta”) serve as a reliable automated red-flag for instructor attention? Specifically, do teams with a large Delta between the simple average and the iterative or normalisation models correspond to teams exhibiting suspicious Git contribution patterns (high Gini Index) or verified free-riding in the historical record?

This question establishes the practical utility of running multiple models in parallel. If a team’s IWF distributions are consistent across all four models, the instructor can treat the scores as reliable. If one model produces a substantially different rank ordering, that divergence is an informative signal that warrants human investigation.

4. **RQ4 — Qualitative Red-Flagging via Journal Sentiment:** To what extent do discrepancies between a student’s received peer scores and the sentiment of their periodic journal entries identify cases of involuntary non-contribution that numerical models alone cannot detect?

This question targets the blind spot of all purely numerical models: they cannot distinguish a student who chose not to contribute from one who was socially excluded from contributing. Hall and Buzwell [2] document cases in which students’ written accounts reveal exclusion dynamics that peer scores mask. A student who writes negatively about team cohesion while receiving high peer scores presents a different risk profile from one whose negative writing aligns with low scores. This question asks whether that divergence is a reliable classifier.

3 Research Methods

3.1 Data Sources

This study will draw on three distinct data sources, each providing a different lens on individual contribution.

Historical peer-score data. Anonymised peer-assessment scores from prior offerings of the University of Auckland’s capstone courses will form the primary dataset. Each record consists of the peer-score matrix for one team: an $N \times N$ matrix where entry (i, j) is the score awarded by student j to student i . The dataset will be preprocessed to remove teams with incomplete submissions, and students who withdrew from the course will be excluded.

Periodic journal entries. Where available, anonymised periodic reflective journals submitted by students will be used as the qualitative data stream. These journals are the source material for the sentiment analysis pipeline described in Section 3.5.

Git repository metrics. Git commit logs from student repositories will be mined to extract per-student contribution metrics: number of commits, lines of code changed (churn), number of completed user stories (issues closed via merged pull requests), and commit frequency over time. Following Hundhausen et al. [5] and Hamer et al. [3], these metrics will be used to compute relative share values for each team member and to calculate per-team Gini Index values as an independent, objective estimate of contribution inequality.

3.2 Ethical Considerations

This study analyses anonymised historical data collected as part of standard course operations. All peer-score matrices, journal entries, and Git logs will be de-identified before the researcher receives access, with student identifiers replaced by opaque tokens. No re-identification will be attempted at any stage.

Reflective journal entries require particular care, as they may contain sensitive disclosures about interpersonal conflict, mental health, or experiences of exclusion. The sentiment analysis pipeline processes these entries computationally; individual journal content will not be reproduced or quoted in any publication. Only aggregate statistical patterns (e.g., mean sentiment trajectories and their correlation with IWF scores) will be reported. Formal ethics approval will be sought from the University of Auckland Human Participants Ethics Committee prior to data access.

3.3 Grading Engine Implementation

The core technical deliverable is a Python-based Grading Engine that accepts a peer-score matrix as input and returns IWF vectors under four computational models. The following descriptions are intended to be sufficient for independent reimplementations.

Model 1 — Baseline (Simple Average). The standard Kaufman et al. [6] peer-rating formula. Given a team of N students and the $N \times N$ score matrix S where s_{ji} is the score awarded to student i by peer j :

1. Discard self-scores: set $s_{ii} = \text{null}$ for all i .

2. Compute each student's IWF as the mean of the scores they received from peers: $\text{IWF}_i = \frac{1}{N-1} \sum_{j \neq i} s_{ji}$.
3. Compute the individual grade as $g_i = G_{\text{team}} \times \frac{\text{IWF}_i}{10}$, where G_{team} is the shared team grade and 10 is the expected score under equal contribution.

This serves as the comparison baseline against which the performance of all other models is measured.

Model 2 — WebPA (Normalisation). The WebPA normalisation algorithm [8] scales scores so that the sum of weighted individual grades equals the team grade.

1. For each student i , compute the total score received from all peers (excluding self): $T_i = \sum_{j \neq i} s_{ji}$.
2. Compute the team mean of total received scores: $\bar{T} = \frac{1}{N} \sum_{i=1}^N T_i$.
3. Compute each student's Peer Assessment Factor: $\text{PAF}_i = \frac{T_i}{\bar{T}}$.
4. Compute the individual grade: $g_i = G_{\text{team}} \times \text{PAF}_i$.

Grade neutrality is preserved: $\sum_i g_i = N \times G_{\text{team}}$. WebPA is immune to the zero-self exploit because the PAF is computed from received peer scores only; a student's allocation of points to themselves does not enter the calculation. However, if all team members submit identical distributions, every PAF equals 1.0 and the mechanism fails to differentiate.

Model 3 — PeerRank (Iterative Graph). Walsh's [12] fixed-point algorithm, adapted here for the IWF context. Let the normalised score matrix A have entries $a_{ji} = s_{ji} / \sum_k s_{jk}$, representing the fraction of peer j 's total budget allocated to student i . The algorithm proceeds as follows:

1. Initialise grades to the simple average: $X_i^{(0)} = \frac{1}{N-1} \sum_{j \neq i} a_{ji}$ for all i , then normalise so that $\sum_i X_i^{(0)} = 1$.
2. At each iteration t , update grades:

$$X_i^{(t+1)} = (1 - \alpha) X_i^{(t)} + \alpha \frac{\sum_{j \neq i} X_j^{(t)} \cdot a_{ji}}{\sum_{j \neq i} X_j^{(t)}} \quad (2)$$

where $\alpha \in (0, 1)$ is the learning rate (set to $\alpha = 0.1$ following Walsh). The grade assigned by peer j is weighted by j 's own current grade $X_j^{(t)}$: a poorly-rated student's votes are dampened, and a highly-rated student's votes carry more weight.

3. Iterate until the L_1 norm of successive updates falls below a convergence threshold $\epsilon = 10^{-6}$: $\sum_i |X_i^{(t+1)} - X_i^{(t)}| < \epsilon$.
4. Convert the converged grade vector to IWFs by scaling relative to the team mean: $\text{IWF}_i = X_i / \bar{X} \times 10$.

It is noted that PeerRank was designed and validated for settings where students grade independent individual submissions; its application here to peer-assessed team contributions represents an exploratory extension. RQ2 specifically investigates whether the algorithm's convergence properties remain stable in this setting.

Model 4 — PeerHITS (Dual-Score Iterative Graph). A novel peer assessment variant proposed in this research, inspired by Kleinberg’s HITS algorithm [7]. Where PeerRank assigns each student a single grade that simultaneously measures contribution and weights their votes, PeerHITS maintains two separate scores per student: an **authority score** a_i , representing the quality of student i ’s contribution as judged by their peers, and a **hub score** h_i , representing the reliability of student i ’s assessments of others.

Formally, let S be the $N \times N$ peer-score matrix where entry s_{ji} is the score awarded to student i by peer j (with self-scores excluded). The algorithm proceeds as follows:

1. Initialise: $a_i^{(0)} = h_i^{(0)} = 1/N$ for all i .
2. At each iteration t , perform the authority update:

$$a_i^{(t+1)} = \sum_{j \neq i} h_j^{(t)} \cdot s_{ji} \quad (3)$$

A student’s authority score is the sum of the scores they received, each weighted by the hub score of the rater. Raters with high hub scores (i.e., those whose assessments align with consensus) contribute more to the authority of the students they rate highly.

3. Perform the hub update:

$$h_i^{(t+1)} = \sum_{j \neq i} a_j^{(t)} \cdot s_{ij} \quad (4)$$

A student’s hub score is the sum of the authority scores of the students they rated, each weighted by the score they gave. A student who assigns high scores to genuinely strong contributors receives a high hub score.

4. L_2 -normalise both vectors: $a_i \leftarrow a_i / \|\mathbf{a}\|_2$, $h_i \leftarrow h_i / \|\mathbf{h}\|_2$.
5. Iterate until $\sum_i |a_i^{(t+1)} - a_i^{(t)}| < \epsilon$ with $\epsilon = 10^{-6}$.
6. Derive IWFs from the authority scores, scaled to preserve grade neutrality: $\text{IWF}_i = a_i / \bar{a} \times 10$.

The hub score h_i is retained as a diagnostic signal rather than a grading input: a student with a substantially lower hub score than their peers may be an unreliable rater, warranting instructor attention independently of their authority-derived grade. This separation is the key structural advantage of PeerHITS over PeerRank — it distinguishes *contribution quality* from *assessment quality* rather than conflating both into a single fixed-point.

Note that, like PeerRank, PeerHITS operates on the post-exclusion peer-score matrix and is therefore not inherently resistant to the zero-self exploit. The surplus points injected by coordinated self-zeroing inflate the raw scores before either algorithm processes them. Detecting this exploit requires a pre-processing check (e.g., flagging teams where one or more members allocated zero points to themselves) rather than a property of the iterative model itself.

PeerHITS has not previously been applied to the IWF context; this work constitutes its first empirical evaluation in a capstone team assessment setting.

3.4 Synthetic Attack Simulation

Each of the four attack vectors simulated in this section corresponds to a documented pattern of strategic behaviour observed empirically in peer-assessed group work, rather than a hypothetical worst case. The attacks are motivated as follows.

Uniform inflation attack. Song et al. [11] identify *pervasive collusion* as one of the two principal failure modes in peer assessment: a pattern in which reviewers assign top scores to all submissions they evaluate, without any explicit coordination. In their dataset spanning multiple disciplines, nearly 10% of all rating-based reviews were submitted by pervasive colluders, with the worst-case grade inflation reaching 5–10 points on a 100-point scale. Critically, Song et al. also find that pervasive collusion exerts a “peer pressure” effect on honest reviewers, who become more lenient when surrounded by inflated scores — making the systemic impact larger than the colluding minority alone would suggest. Hooshangi et al. [4] confirm this pattern in capstone settings through instructor interviews, with respondents explicitly identifying tacit mutual-inflation agreements as the default failure mode in cohesive teams. The uniform inflation attack directly simulates this scenario: all N members of a team submit uniform scores of $10/N$ each, equalising all IWFs and rendering the peer-assessment mechanism entirely ineffective.

Zero-self attack. Song et al. [11] further identify *small-circle collusion*, in which a coordinated subset of students exchange inflated scores with one another. In the IWF model, a fully coordinated team has an additional exploit unavailable in the individual-submission settings Song et al. studied: because self-scores are silently discarded but the denominator of the IWF remains fixed at $N - 1$, a student who awards themselves zero effectively injects surplus points into the pool available to teammates. When the whole team coordinates this manoeuvre, every member receives a grade uplift with no corresponding increase in contribution. This attack tests whether any model inadvertently rewards or is destabilised by this structural exploit, which represents the most aggressive form of small-circle collusion possible within the IWF scoring rules.

Targeted down-vote attack. Hall and Buzwell [2] document through thematic analysis of 205 students that status hierarchies crystallise early in group projects, and that low-status members are sometimes deliberately excluded from contribution by higher-status teammates who judge their involvement harmful to the team’s grade. Vernon (2008, cited in [2]) describes how students who fail to establish themselves in the first weeks find themselves in a position of “involuntary free-riders” through social dynamics rather than individual choice — and are subsequently assessed as such by peers who have acquired ownership of the project. The targeted down-vote attack simulates the grading expression of this coordinated exclusion: the “victim” receives a score of 0 from all other team members, while the remaining members award each other equal shares.

Single outlier attack. Piech et al. [10] demonstrate empirically, analysing 63,199 peer grades from Coursera’s HCI course, that individual graders vary substantially and systematically in both bias and reliability — some consistently inflate, others consistently deflate, independently of submission quality. In a capstone team of four to six students, a single unreliable rater represents 20–25% of the grading graph; their influence on the IWF is proportionally far greater than in the large MOOC cohorts Piech et al. studied. This attack replaces one student’s scores with a random permutation of the full point budget, modelling a unilateral rater acting either in bad faith or from genuine ignorance. It specifically tests PeerRank’s convergence stability in small graphs (RQ2).

For each attack vector, the *Attack Delta* is computed as the mean absolute difference between the IWF vector under the attack and the IWF vector computed from the unmodified scores. A model is considered more robust if its Attack Delta is smaller.

3.5 Qualitative Red-Flag Pipeline

Periodic journal entries will be processed through a two-stage sentiment analysis pipeline. In the first stage, VADER (Valence Aware Dictionary and sEntiment Reasoner), available through NLTK, will be applied to each journal entry to produce a compound sentiment score in the range $[-1, 1]$. A rolling mean will then be computed for each student across the course duration, producing a sentiment trajectory.

A known limitation of VADER is that it was designed for short, informal social media text and may underperform on the longer, more hedged prose typical of reflective academic journals. Students expressing frustration through qualified or indirect language (e.g., “the team dynamic has been somewhat challenging”) may not register as negative under VADER’s lexicon-based scoring. If initial results indicate poor sensitivity, a transformer-based alternative (e.g., a fine-tuned BERT model or zero-shot classification via a large language model) will be evaluated as a replacement. The choice of sentiment tool will be reported alongside results for RQ4.

In the second stage, each student’s sentiment trajectory will be compared against their received IWF trajectory (computed from weekly peer scores where available, or the single end-of-project IWF otherwise). Four cases are of theoretical interest, following the framework of Hall and Buzwell [2]:

1. **Aligned positive:** High IWF, positive sentiment. Expected for a high-performing, well-integrated team member.
2. **Aligned negative:** Low IWF, negative sentiment. Expected for a voluntary free-rider who is both assessed poorly and disengaged.
3. **Divergent — suppressed distress:** Low IWF, positive or neutral sentiment. May indicate a student who has accepted a reduced role or has underreported difficulties. Lower priority for intervention.
4. **Divergent — masked exclusion:** High IWF but consistently negative sentiment about teamwork and collaboration. This is the primary target of RQ4: a student who writes negatively about the team dynamic but receives inflated peer scores is a candidate for involuntary free-riding or social exclusion. Such cases will be flagged for instructor review.

Cases in category 4 will be identified by applying a threshold to the difference between the standardised IWF z-score and the standardised sentiment z-score. The threshold will be calibrated against any ground-truth information available in the historical record (e.g., documented complaints or grade appeals).

3.6 Evaluation Framework

The performance of each model will be assessed on three criteria:

- **Rank Reversal Rate:** The proportion of teams in the historical dataset where an advanced model produces a substantially different rank ordering of students (defined as at least one pair of students whose IWF order is reversed by more than a threshold $\delta = 0.15$) compared to the simple average. A higher rate indicates that the model is detecting signal that the baseline misses.
- **Attack Robustness:** The Attack Delta (defined above) under each of the four synthetic attack vectors.

- **Git Alignment:** The Pearson correlation between each model’s IWF vector and the corresponding relative commit share vector from Git metrics, computed across all teams with available repository data. A higher correlation is evidence that the model’s IWF more accurately reflects observable contribution than the baseline.

3.7 Threats to Validity

Several threats to validity should be acknowledged.

Ground-truth limitations. No direct ground truth for individual contribution exists in the historical dataset. Git metrics serve as the primary external validation signal, but they are an imperfect proxy: Hundhausen et al. [5] note that commit counts and lines changed do not capture leadership contributions, pair programming, architectural design, or other non-code activities. A student who plays a critical coordination role but writes relatively little code will appear as a low contributor in Git metrics. The Git Alignment criterion should therefore be interpreted as a necessary but not sufficient condition for model accuracy; a model that correlates poorly with Git metrics is likely inaccurate, but a model that correlates well is not guaranteed to be capturing the full picture.

Small-graph effects. All models are evaluated on teams of 4–6 students, where the influence of any single rater is substantial. Results may not generalise to larger cohorts, and convergence properties established in the literature for larger groups may not hold.

Sentiment tool sensitivity. As noted in Section 3.5, VADER may lack the sensitivity required for hedged academic prose. The findings of RQ4 are contingent on the sentiment tool’s ability to detect meaningful affect variation in journal entries.

3.8 Tools

- **Python 3.x:** Core implementation language. Jupyter notebooks will be used for interactive development, incremental analysis, and reproducible documentation of experimental results.
- **Polars / Pandas / PyArrow:** Data ingestion, cleaning, and manipulation. Polars will be used for performance-critical transformations on the full historical dataset; Pandas for ad-hoc exploratory analysis; and PyArrow for efficient columnar serialisation of intermediate results.
- **NumPy:** Matrix operations underpinning the Grading Engine, including the IWF computations and the PeerRank fixed-point iteration.
- **NetworkX:** Graph construction and analysis for the PeerRank and PeerHITS iterative models. Each team’s peer-score matrix will be represented as a weighted directed graph, with NetworkX used to inspect convergence properties, compute node centrality, and visualise the grading network structure.
- **scikit-learn:** Machine learning utilities for the qualitative red-flag pipeline, including z-score standardisation, threshold calibration, and classification of the four sentiment-vs-IWF divergence cases described in Section 3.5.
- **NLTK (VADER):** Sentiment analysis of periodic journal entries, producing compound sentiment scores in the range $[-1, 1]$ for each submission.

- **Pydantic:** Data validation and schema enforcement for peer-score matrices and journal records ingested from the historical dataset, ensuring type safety and catching malformed inputs before they propagate into the Grading Engine.
- **Plotly:** Interactive visualisation of IWF distributions, Gini Index inequality plots, Delta-Analysis heatmaps, and sentiment-vs-IWF divergence charts. Interactive plots are particularly suited to the instructor-facing outputs of this project, where the ability to drill down into individual teams and students would be valuable in practice.
- **GitHub GraphQL API:** Extraction of Git contribution metrics (commits, lines changed, issues closed) from student repositories.

4 Timeline

Figure 1 presents the project timeline as a Gantt chart. The project spans 15 weeks and is organised into four parallel workstreams: Literature and Data, Implementation, Analysis, and Writing. Implementation and Analysis overlap deliberately: the Grading Engine is validated against analytical examples before being applied to the historical dataset, and dissertation writing begins during the final analysis phase rather than after it, to avoid a bottleneck in the last two weeks.

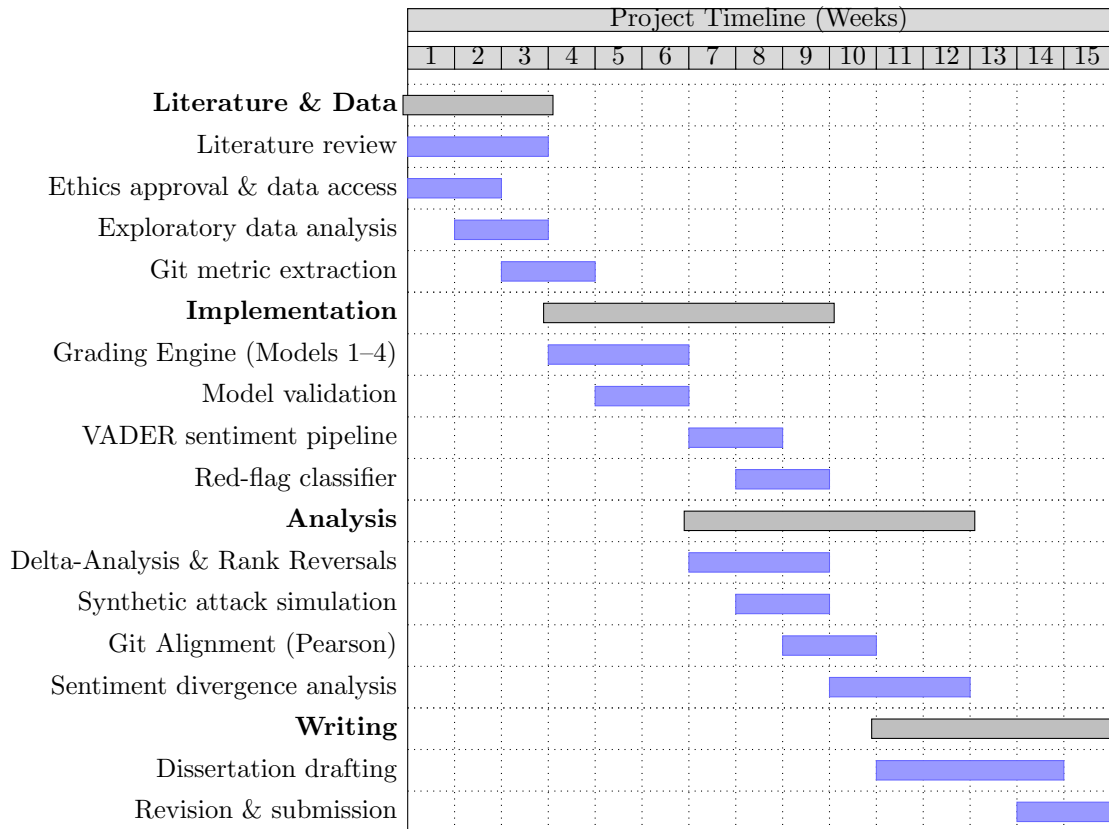


Figure 1: Project timeline as a Gantt chart.

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